Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. What function does the discriminator serve within a Generative Adversarial Network (GAN)? 1 / 1 point

- It introduces noise into the input data.
- It produces synthetic data.
- It discerns between authentic and fabricated data.
- It refines the loss function.

Nice work

Correct! The discriminator's primary role is to differentiate between genuine data and data generated by the generator.

2. What is the principal goal of employing synthetic data in artificial intelligence applications? 1 / 1 point

- To accelerate the model training process
- To completely substitute real data
- To streamline the model's architecture
- To enhance model efficacy by expanding the training dataset

Nice work

Correct! Synthetic data is primarily used to augment the training set, thereby boosting the model's performance.

3. Consider a scenario where a company's image classification system is frequently misled by subtle, imperceptible changes in the input images. To address this issue, the company decides to implement defense strategies against adversarial attacks. Which of the following approaches would be most effective in improving the robustness of their system? 1 / 1 point

- Employing adversarial training to include adversarial examples in the training process
- Implementing gradient masking to obscure gradient information
- Minimizing the size of the training dataset
- Utilizing higher learning rates during training

Nice work

Correct! Adversarial training helps to make the model more resilient by exposing it to adversarial examples during training.

4. Which method is a prevalent strategy for conducting adversarial attacks on neural networks? 1 / 1 point

- Expanding the size of the training dataset
- Removing layers from the neural network
- Introducing perturbations to input data
- Modifying the network's activation functions

Nice work

Correct! Perturbing input data is a common technique used to trick neural networks into misclassifying inputs.

5. Which of the following statements accurately describe Generative Adversarial Networks (GANs)? 1 / 1 point

- [ ] They fall under the category of reinforcement learning.
- [x] They incorporate both a generator and a discriminator.

Nice work

Correct! GANs consist of two main components: a generator, which creates synthetic data, and a discriminator, which evaluates it.

- [x] They are utilized for the creation of synthetic data.

Nice work

Correct! GANs are extensively used to generate synthetic data for various applications.

- [ ] They operate independently of any training data.